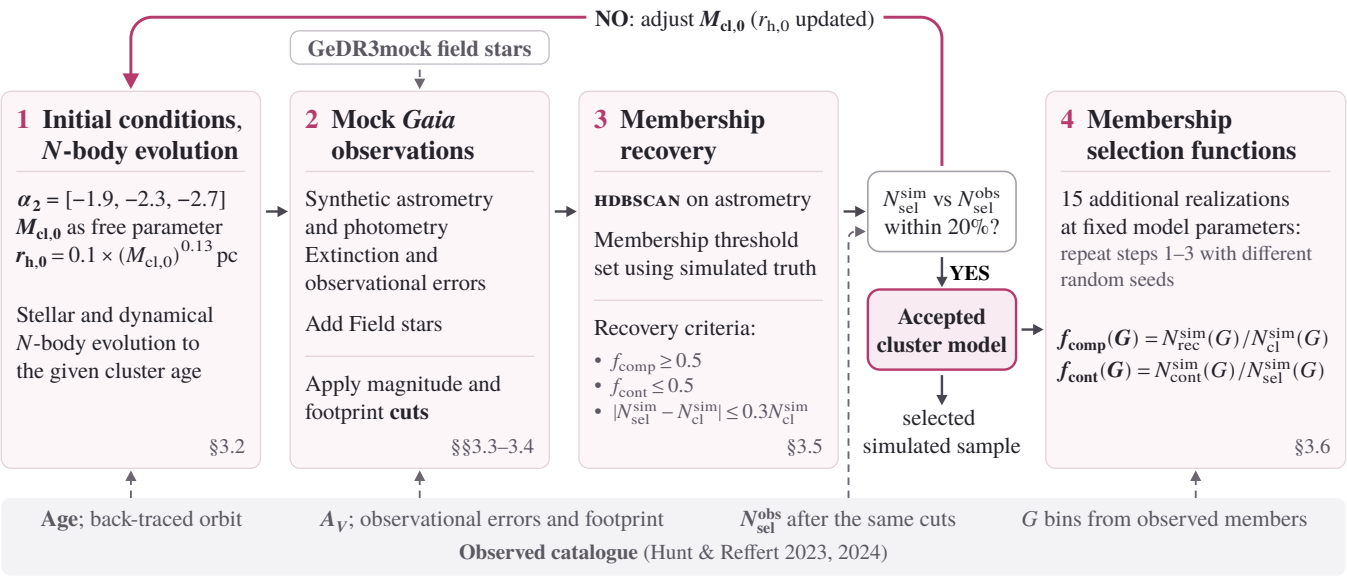
NO: adjust $M_{cl,0}$ ($r_{h,0}$ updated)
GeDR3mock field stars
1 Initial conditions, N-body evolution
$\alpha_2 = [-1.9, -2.3, -2.7]$
$M_{cl,0}$ as free parameter
$r_{h,0} = 0.1 \times (M_{cl,0})^{0.13}$ pc
Stellar and dynamical N-body evolution to the given cluster age
§3.2
2 Mock Gaia observations
Synthetic astrometry and photometry
Extinction and observational errors
Add Field stars
Apply magnitude and footprint cuts
§§3.3–3.4
3 Membership recovery
HDBSCAN on astrometry
Membership threshold set using simulated truth
Recovery criteria:
• $f_{comp} \geq 0.5$
• $f_{cont} \leq 0.5$
• $|N_{sel}^{sim} - N_{cl}^{sim}| \leq 0.3 N_{cl}^{sim}$
§3.5
N_{sel}^{sim} vs N_{sel}^{obs} within 20%?
YES
Accepted cluster model
selected simulated sample
4 Membership selection functions
15 additional realizations at fixed model parameters:
repeat steps 1–3 with different random seeds
$f_{comp}(G) = N_{rec}^{sim}(G) / N_{cl}^{sim}(G)$
$f_{cont}(G) = N_{cont}^{sim}(G) / N_{sel}^{sim}(G)$
§3.6
Age; back-traced orbit
A_V; observational errors and footprint
N_{sel}^{obs} after the same cuts
G bins from observed members
Observed catalogue (Hunt & Reffert 2023, 2024)